



ME 101 INTRODUCTION TO MECHANICAL ENGINEERING

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FALL 2025

- Welcome to the Fall 2025 semester! All students are expected to enroll to the course at the YULEARN (moodle) website. Enrollment key for the course TBA.
- Students must obtain a @std.yeditepe.edu.tr extension email address for signing up to YULEARN. To get your email address send an email containing your student number, name, surname to labadmin@cse.yeditepe.edu.tr.

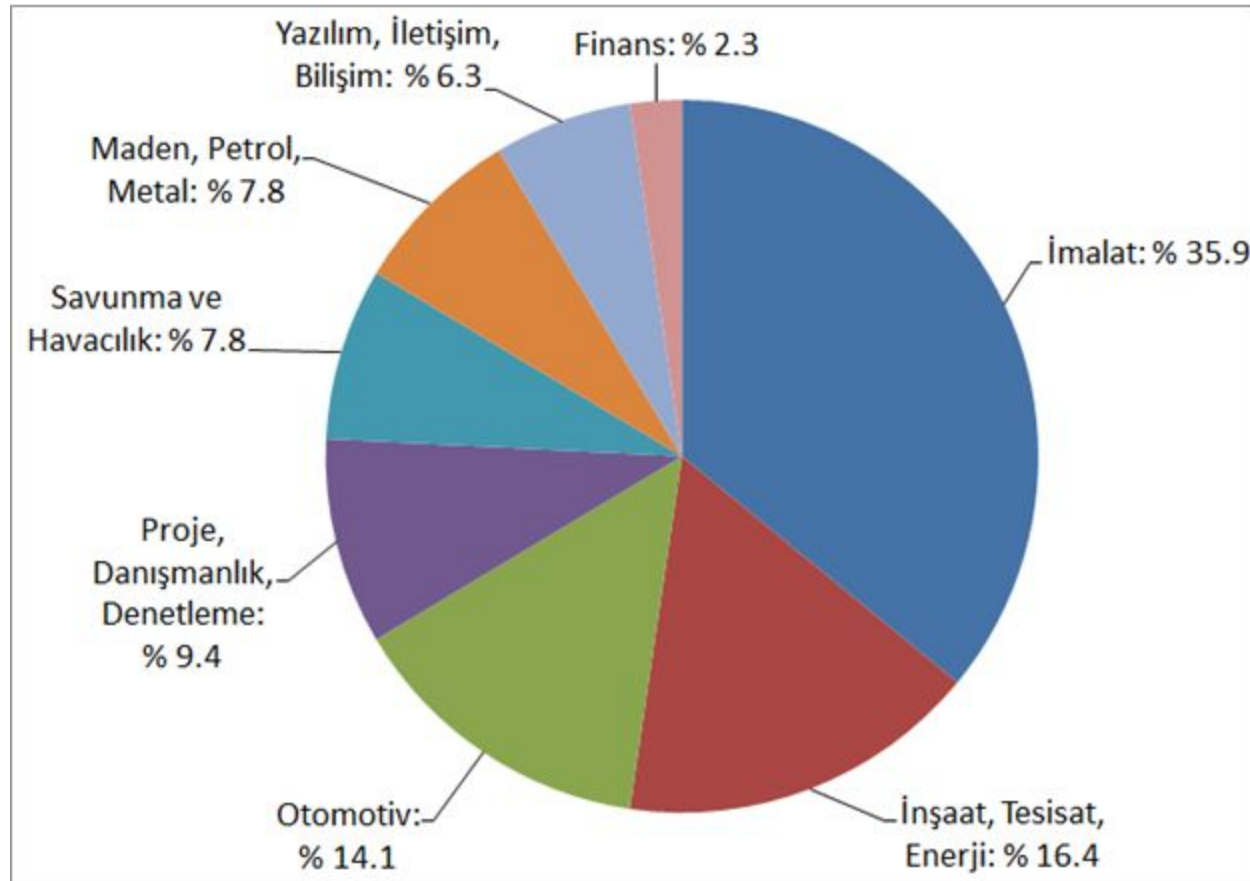
OUR DEPARTMENT

- Mechanical Engineering Department was founded in 2001
- 11 instructors (öğretim elemanı; does not necessarily imply rank), 6 RAs
- Currently ~500 undergraduate students
- >1000 graduates* till now
 - First graduates in 2005
- 6 laboratories
- One workshop (atölye)

Ranks in academia:

- Instructor (öğretim görevlisi)
- Assistant professor (Dr. Ogr. Uyesi)
- Associate professor (doçent)
- Professor (“full professor”) (profesör)

OUR GRADUATES (Based on a survey of 200 grads)



OUR FACULTY

- Founded in 1996 (when 7tepe was first founded)
- Faculties are responsible for **undergraduate education (lisans eğitimi)**
- The name of our faculty:
 - Until 26.8.2014:
→ MÜHENDİSLİK ve MİMARLIK FAKÜLTESİ
 - After 26.8.2014:
→ MÜHENDİSLİK FAKÜLTESİ
 - After 2025:
MÜHENDİSLİK ve DOĞA BİLİMLERİ FAKÜLTESİ
- ~3500 undergraduate students in 10 departments
(excluding the Preparatory School)

FACULTIES AT THE UNIVERSITY

- There are 13 faculties at Yeditepe

→ Some of the faculties:

- Architecture
- Fine Arts (Güzel Sanatlar),
- Law (Hukuk),
- Medicine (Tıp),
- Communication (İletişim),
- etc.

INSTITUTES (ENSTİTÜLER)

- Institutes are responsible for **graduate education (lisansüstü eğitim)**: masters level(yüksek lisans) and doctoral (PhD) level
- There is one institute at Yeditepe
 - Institute of Graduate Education (Yüksek Lisans Enstitüsü)

ADMINISTRATORS

- Head of a department: chairman, chairperson, chair (bölüm başkanı)
- Head of a faculty: dean (dekan)
 - **Faculty Executive Board (Fakülte Yönetim Kurulu):**
authority to decide about daily issues about students
→ Course transfers (muafiyet / ders sayma), semesterly leaves of absence (kayıt dondurma), graduations (mezuniyet), etc.
- Head of an institute: institute director (enstitü müdürü)
- Head of a university: rector (Europe),
president (USA), rektör (Turkish)
- Vakıf üniversitelerinde en üst düzeydeki yetkili birim / kişi:
Mütevelli Heyeti / Mütevelli Heyet Bşk.

WEIGHT OF A COURSE

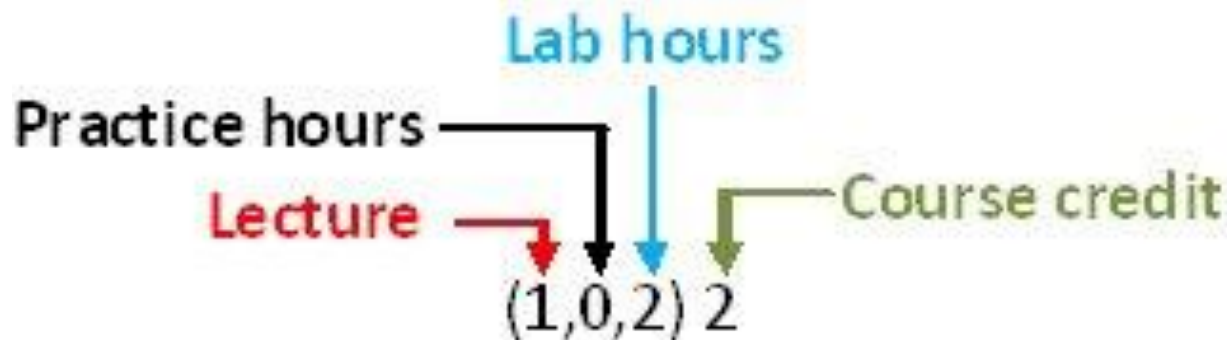
Needed: a measure of the weight of a course

- One **credit-hour** (or, one **credit**) :
one **contact hour** (50 min) per week.
→ **contact hour:** 50 min of lecture **or**
100 min of lab/practice work
- ECTS : **E**uropean **C**redit **T**ransfer
and Accumulation **S**ystem
(AKTS: **A**vrupa **K**redi **T**ransfer **S**istemi)
→ More complicated:
contact hours+time spent outside classroom, ...

Credit Structure of ME 101

- (1,0,2) 2 → → A two credit-hour course

Shortly : A two credit course



- ECTS of ME 101 : 7
- The **total** number of **ECTS** credits is **240** for all four-year undergraduate programs.

CONDUCT OF THE COURSE

There will be

- one hour of lecture and
- two hours of laboratory (“lab”) work

every week.

- There will be a joint section for lectures and five sections for the lab sessions.

CONDUCT OF THE COURSE

- Rule: at least 80 % attendance is required.
⇒ **Don't miss more than 20 %.**
- You **cannot miss more than two lab sessions.**
- Miss more than 20 % total or more than 2 labs
⇒ ⇒ ⇒
 - you get FA,
 - you cannot take the final exam,
 - you have no right to the makeup exam (bütünleme)⇒ ⇒ ⇒ **(You fail the course)**

CONDUCT OF COURSES IN ME DEPARTMENT

- **Do not sign** the attendance sheet **on behalf of an absent friend**
- Do not be late for the class
⇒ if the class-room door is closed, **knock to enter**

IN-CLASS BEHAVIOR IN ME DEPARTMENT

- Bring the textbook to the class
- Make it a habit to **take notes during the lecture**

⇒ **Bring to class**

- a notebook,
- pen, pencil, eraser
- a calculator
- a mini ruler

⇒ Carry these in a small **backpack**

IN-CLASS BEHAVIOR IN ME DEPARTMENT

- **No cell phone use.** Make sure your phone is
 - turned off, or
 - in silent mode.
- No exchange of pencils, erasers, calculators, etc. **in exams. Bring your own!**
- **Do not** attempt to **cheat** in exams:

HOMEWORK RULES IN ME DEPARTMENT

- Use size A4 paper not torn from a notebook.
- **Mark the margins** of the paper on all four sides and **do not write in the margins** (a couple of centimeters on all sides)
- Do not use pens. **Use pencils!**
- The idea behind homework is for you to learn the subject matter. If you **copy homework** from your friends, **the only person you hurt is yourself.**
- If your homework is a copy, you will get zero and disciplinary action may be taken against you
- Turn in your homework in due time. Late ones will either not be accepted or they will be marked down.

- “Engineer” : ingeniosus (Latince), ingenious (İngilizce)
 - a skillful and smart person who invents or builds a clever tool/equipment/machine (becerikli, zeki, hünerli, akıllıca icat ya da imalat yapan)
- “Engine”: a clever and useful equipment (akıllıca yapılmış yararlı aygıt)

ENGINEERING

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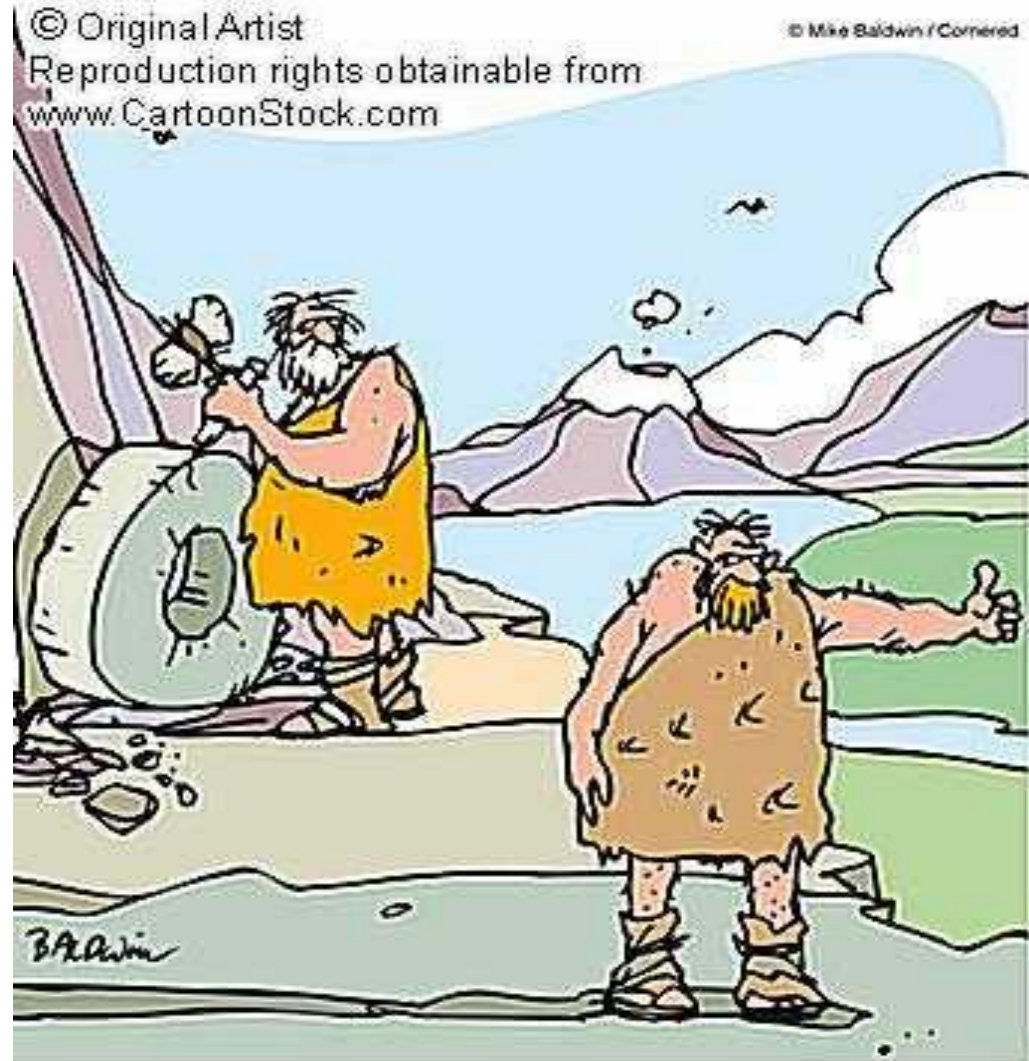
"Art, art, art - When are we going to get some engineers?"

ENGINEERING = MÜHENDİSLİK

- Hendese = geometri
- From the word hendese → mühendis

ENGINEERING

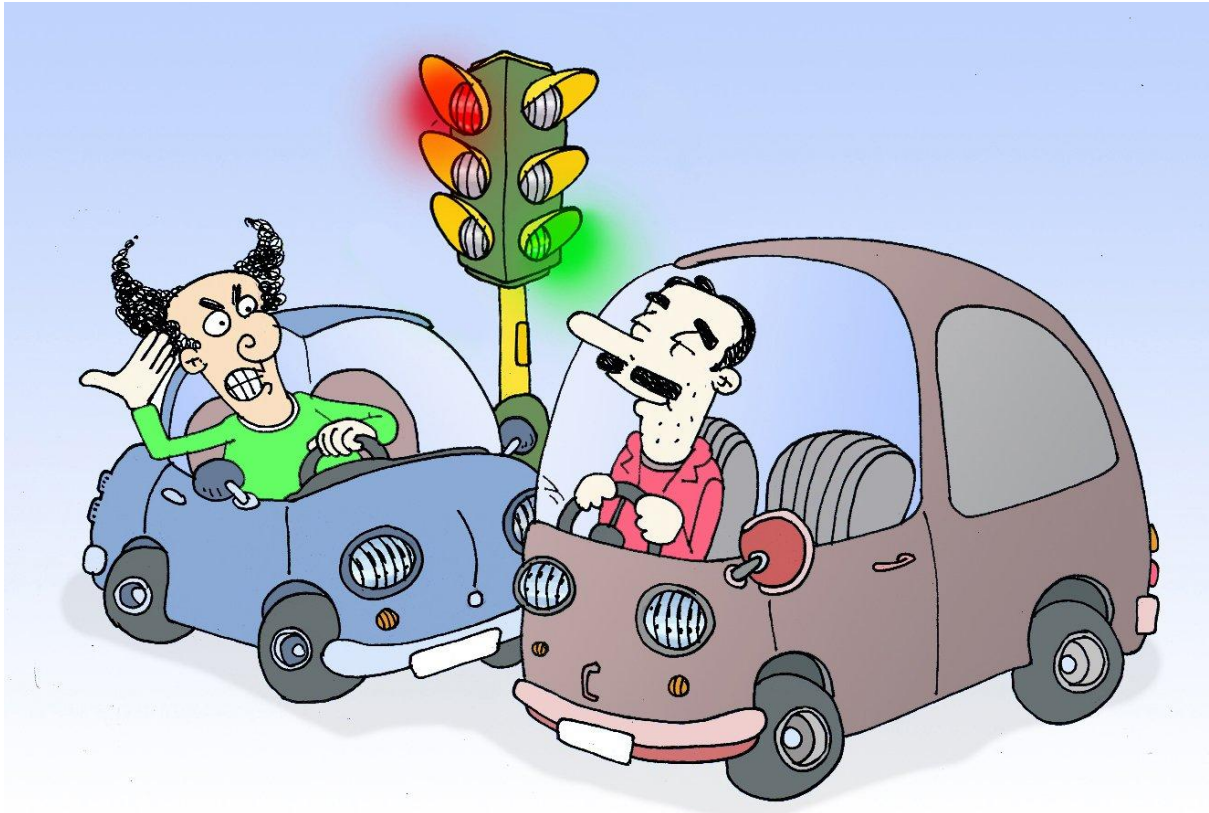
Ancient times;
~ 5000 years
ago



- Engineers apply the theories and principles of **science and mathematics** to find economical solutions to technical problems.
- Mühendislik: **fen bilimleri ve matematiğin**, bir gereksinimi karşılayacak şekilde uygulanarak bir sistem, süreç, aygıt tasarlanıp imal edilmesi.

- Men had to be “engineers:
→ hunting, defense, tool making, ...
- “Engineering” began in ancient times.
→ clever inventions:
wedge, lever, wheel, pulley, ...

- The diplomas that we grant must have substance (Verdiğimiz diplomanın altı dolu olmalı)



- Each graduate is an ambassador of our department; a representative of our future graduates.
- It is **hard to gain** a reputation of “a department with high quality graduates”, but **very easy to loose** it.
- We have to maintain a certain level of quality.
- You **must deserve to graduate**.

YOU MUST WORK HARD!

SUCCESS/FAILURE AS A STUDENT

A few reasons for failure
in your first year in the department:

- **TYT-AET** fatigue $\Rightarrow$ laziness
- Not knowing the chosen field (ME) closely.
- “I already know math and physics”.
- To have a weak foundation (temeli zayıf olmak).
-

SUCCESS/FAILURE AS A STUDENT

- A common scenario of repeated failure (“Ben zaten matematik, fizik biliyorum”.)
 - İki arkadaşın sohbeti (yer: Mühendislik kantini)
 - A- Sınıfta hoca türev, limit filan anlattı.
 - B- Dersaneden bildiğim konular; çalışmama gerek yok, biraz baksam yeter.

SUCCESS/FAILURE AS A STUDENT

→ → Just because some math or physics terms are familiar to you, **do not think that you know the subject!!!**

- Unfortunately, today high school and “dersane” education teaches you **how to solve a problem** in 1-2 minutes.
- As a result, in most cases you do not understand **why you solve the problem that way**.
- Engineering requires you to **know the WHY!**

SUCCESS/FAILURE AS A STUDENT

→ 16 hafta sonra

A- Calculus'ten ne aldın?

B- F!!!

A) Hani biliyordun dersi.

B) Valla ben de anlamadım

SUCCESS/FAILURE AS A STUDENT

- University physics and math education:
 - Aims to teach you the “**why of phenomena**”.
 - Aims to **train your brain** so that you learn to think like an engineer.
 - **Analytical thinking, ability to reason.**
 - So that you can analyze problems and find solutions in your future career.
- **In your future career as an engineer:**
 - No one will give you 5 possible solutions and ask you to choose one.
 - You will be asked to **figure out** (find) **a solution on your own**, or even, to **find a non-existent solution**.

SUCCESS/FAILURE AS A STUDENT

→ Bir dönem sonra (yer: çimler) :

A- Calculus nasıl gidiyor?

B- Bir sürü soru çözdüm, AA hedefliyorum.

→ 14 hafta sonra:

A) Calculus ne geldi?

B) F!!!

SUCCESS/FAILURE AS A STUDENT

- Gaining the ability to think like an engineer:
 - Not possible by **ONLY** trying to solve (**OR**, to “read” the solutions of) as many problems as you can!
 - **OR**, by listening to a friend solving problems
- You **must understand** the underlying principles (**the why**).
 - ⇒ **must study the subject matter** in your text book (**kitaptan konuyu çalışmalısınız!!!**)
 - Then attempt to solve problems...

SUCCESS/FAILURE AS A STUDENT

→ İki dönem sonra (yer: Kadıköy'de bir kafe)

A- Calculus çalışıyor musun?

B- Sabah akşam soru çözüyorum.

→ 14 hafta sonra:

A) Calculus notun?

B) F!!

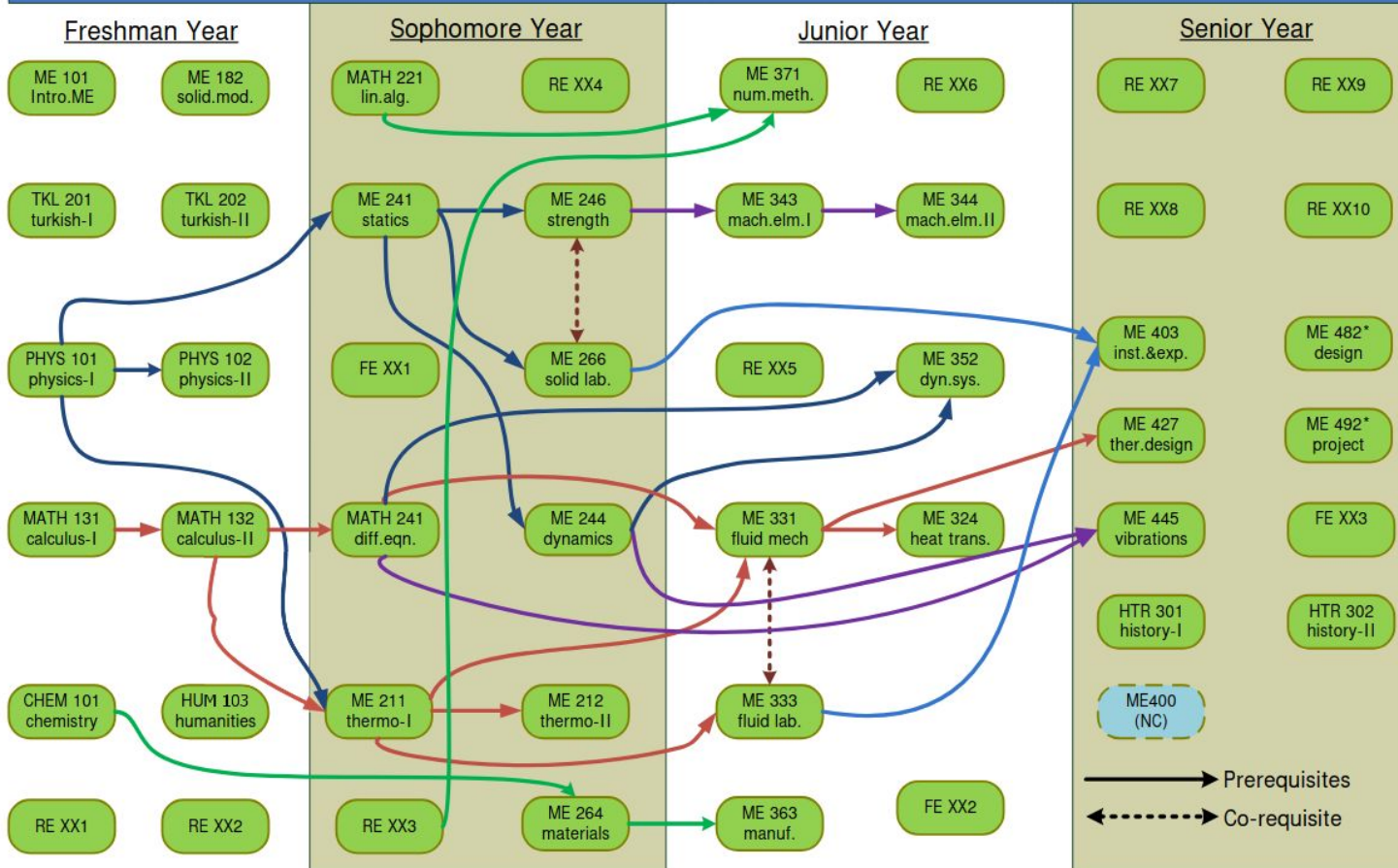
SUCCESS/FAILURE AS A STUDENT

Some reasons that **will get you in trouble later:**

- **Prerequisite chain (önkoşul zinciri):**
→ (You cannot build a structure without a **foundation**)
- Have barely passed MATH and PHYSICS courses (“hasbelkader” geçmiş olmak)
→ **lack of adequate knowledge**
- Got DD and DC from high-credit courses
→ **CGPA problem (ortalama sıkıntısı)**

PREREQUISITE CHAIN

MECHANICAL ENGINEERING DEPARTMENT CURRICULUM AND PREREQUISITES



* Sadece mezuniyet durumundaki öğrenciler ME 482 ve ME 492 derslerini alabilir

* Only students who are in their final semester for potential graduation can take ME 482 and ME 492.

01.09.2024

EXAMPLES OF PREREQUISITE CHAINS

- **MATH 131 → MATH 132 → MATH 241 → ME 331**
ME 352
ME 445
- **PHYS 101 → ME 241 → ME 246 → ME 343**
→ ME 344
- **MATH 131 → MATH 132 → ME 211 → ME 331**
→ ME 324

CALCULATION OF GPA (Grade Point Average)

Each letter grade has a corresponding **multiplier**:
AA→4.00, BA→3.50, BB→3.00, ..., DD→1.00, FF→0

- You get DC from a 4-credit course: **quality points** you earn is $1.50 \times 4 = 6.00$
- Add all quality points you have earned in a semester
- Divide the total quality points (36) by the total number of credits (20) you took that semester
⇒ you find your **semesterly GPA** (say, 1.80),
- Do this for all the semesters till now to find **CGPA**
(Cumulative GPA = Cumulative Grade Point Average)

(Genel Not Ortalaması = **GNO** in Turkish)

SUCCESS/FAILURE AS A STUDENT

The price you pay for passing high-credit courses with low grades:

Example:

Chem 101 (**4**) : DD

MATH 151 (**4**) : DD

PHYS 101 (**4**) : DD

ME 101 (2) : BB

AFE 131 (3) : BB

TKL 201 (3) : BB

$$\Rightarrow \text{GPA} = 36/20 = 1.80$$

(If it were a simple average:

3 DDs and 3 BBs $\Rightarrow$ the average would be CC (2.00)

It is a **weighted average!**

Weighted by the course credits!

SUCCESS/FAILURE AS A STUDENT

Example: A student in his/her **3rd** year
(**NOT “3. year”!**) who has taken
70 credits worth of courses so far.

- Say, s/he has a cumulative GPA of 1.75
- Means she collected a total quality points of
CGPA*completed credit = $1.75 * 70 = 122.5$

SUCCESS/FAILURE AS A STUDENT

Example (continued).

- Currently, s/he takes a course load of 15 credits:
In order to bring his/her **CGPA to 2.00** at the end of the current semester, s/he must accumulate a total of

$$(70+15)*2.00=\mathbf{170} \text{ quality points}$$

→ S/he already has **122.5**

⇒ During the current semester s/he must earn **$170-122.5=47.5$** quality points

⇒ His/her **GPA for the current semester** must be **$47.5/15=3.17$** . (> BB)

MORAL OF THE STORY

Low grades in the first two years
are harder to compensate for
in later years!!!

SUCCESS/FAILURE AS A STUDENT

- Aim a high CGPA; do not just barely pass your courses.
- Toward that aim, study hard beginning in your first year. **4-credit courses** are in the first two years.
- A high CGPA is useful for
 - Getting the most prestigious jobs
 - Getting good scholarships for graduate study
 - For getting achievement scholarship (**başarı bursu**) from Yeditepe.

And, it might make you feel good, proud of yourself

SOME USEFUL ADVICE

- Know the curriculum (“müfredat”) well.
- Know the prerequisites well.
- Learn how to calculate your CGPA.
- Keep an **up-to-date excel file for the courses** you have taken.

The ME Curriculum (course programme)

<https://eng.yeditepe.edu.tr/tr/makine-muhendisligi-bolumu/dersler>

SOME USEFUL ADVICE

- Kayıt
- Ders ekleme bırakma (“add-drop”)
- Dersten çekilme (“withdrawal”)
- Ders saydırma, dilekçe işleri
- Son tarihlere dikkat etmek
- Panoları izlemek

ÖNEMLİ HUSUSLAR ve ÖNERİLER

- Kayıta önce F ve W dersleri
- Çakışma en çok 1 saat: daha fazlasında ders bırakılmak zorunda — PANOLAR

- E-dönüşümdeki (OBS) bilgileriniz doğru olsun: e-posta, cep tel ****

ÖNEMLİ HUSUSLAR ve ÖNERİLER

- Bölümünüzün web sayfasını izleyiniz
DUYURULAR****
- Yönetmelikler:
ÜNİVERSİTE ANA SAYFASI****
- Akademik takvim

ÖNEMLİ HUSUSLAR ve ÖNERİLER

- Sunuş becerileri
- Lisan becerileri

TÜRKÇE ve İNGİLİZCE*****

- Serbest seçmeliler
- KOPYA***